A QUASI-COMPACT REGULAR SCHEME WITH AFFINE DIAGONAL HAS AN AMPLE FAMILY OF INVERTIBLE MODULES

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 3553-3559 and proof span 3561-3581 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement refers to the definition of an ample family of invertible modules in another chapter; the reference is part of the counted statement span and is handled as a navigation pointer.

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Allowed standard prerequisites (editorial).

- A regular scheme is a finite disjoint union of integral schemes, and a normal Noetherian scheme has finitely many irreducible components, cited from the properties chapter. Cited at proof lines 3561-3564. Reduces the statement to the integral case.
- Local rings of a regular scheme are unique factorisation domains, cited from the more-algebra chapter. Cited at proof lines 3569-3570. Makes the complement-of-an-affine-open statement applicable.
- For a quasi-compact scheme with affine diagonal and UFD local rings, the complement of an affine open is the support of an effective Cartier divisor, cited from the same chapter. Cited at proof lines 3570-3572. Produces the divisor whose associated invertible module is the candidate of the ample family.
- The invertible module associated with an effective Cartier divisor and its canonical section, whose nonvanishing locus is the complement of the divisor, cited from the same chapter. Cited at proof lines 3573-3577. Identifies the nonvanishing locus of the canonical section with the chosen affine open, which is the cap condition for an ample family.

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Source reference. Original definition 01WX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2498–2498. Same-chapter reference used in the retained author text at source line(s) 3576, 6928, 9257. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0EGJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18099–18099. Same-chapter reference used in the retained author text at source line(s) 3572. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0AG0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13178–13178. Same-chapter reference used in the retained author text at source line(s) 3149, 3570. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition 0FXR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19891–19891. Same-chapter reference used in the retained author text at source line(s) 3558, 3579. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 033M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4061–4061. Same-chapter reference used in the retained author text at source line(s) 3565. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0567: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6700–6700. Same-chapter reference used in the retained author text at source line(s) 3564. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a quasi-compact, regular scheme with affine diagonal. Then X has an ample family of invertible modules (Morphisms, Definition 0FXR).*

Proof. Observe that X is a finite disjoint union of integral schemes (Properties, Lemmas 0567 and 033M). Thus we may assume that X is integral as well as Noetherian, regular, and having affine diagonal. Let $x \in X$. Choose an affine open neighbourhood $U \subset X$ of x. Since X is integral, U is dense in X. By More on Algebra, Lemma 0AG0 the local rings of X are UFDs. Hence by Lemma 0EGJ we can find an effective Cartier divisor $D \subset X$ whose complement is U. Then the canonical section $s = 1_D$ of $\mathcal{L} = \mathcal{O}_X(D)$, see Definition 01WX, vanishes exactly along D hence $U = X_s$. Thus both conditions in Morphisms, Definition 0FXR hold and we are done. □